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The effect of a lithium niobate heating on the etching rate in SF₆ ICP plasma

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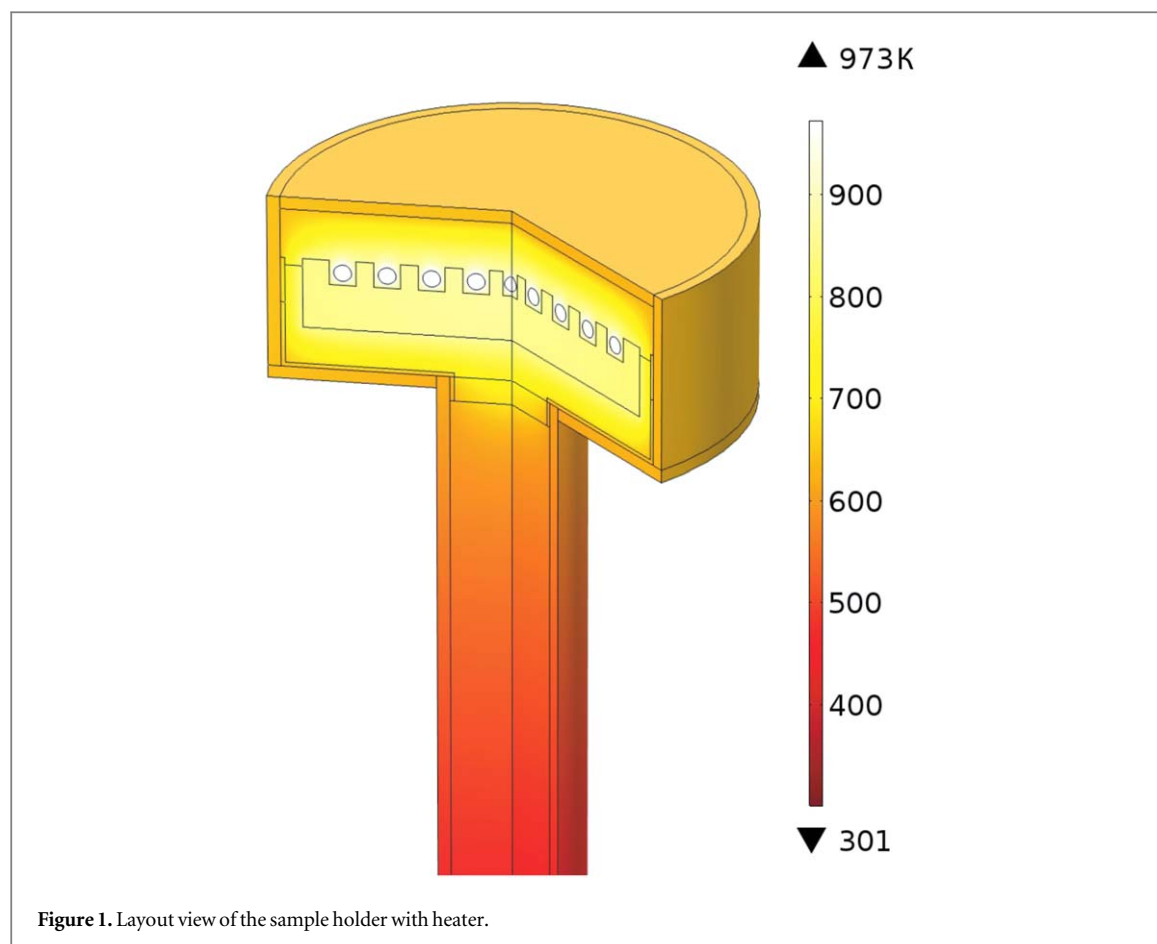
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Abstract

The effect of substrate temperature on the plasma-chemical etching rate of lithium niobate (LiNbO₃) single-crystals in an inductively coupled plasma (ICP) is studied in this paper. we found that the etching rate is a complex function of a substrate temperature with an interval where increases in the temperature lead to a significant growth of the etching rate. comparing the results of LiNbO₃ etching in two different gas mixtures (SF₆/O₂ and SF₆/Ar) showed that the etching process in SF₆/O₂ mixtures provides a higher productivity and a qualitatively better surface roughness. the maximum etching rate achieved during this research was more than 800 nm min⁻¹. The deep linbo₃ etching process (more than 113 μm) using an applied high frequency (HF) source with a power of 700 W was implemented, resulting in an etching rate of more than 420 nm min⁻¹.

1. Introduction

Lithium Niobate (LiNbO₃) single-crystals are widely applied as a material used in a variety of optoelectronic devices. In recent years LiNbO₃ has become a base for different classes of functional and digital integrated optical circuits (IOCs) such as switching matrixes, spectrum analyzers, ultra-high frequency (UHF) phase and amplitude modulators and different physical value sensors [1]. The application of LiNbO₃ as the main material in modern nonlinear optics, electro-optics and acousto-electronics is due to its relatively high values of electrical-, acoustic- and nonlinear-optical coefficients [2–4]. Plasma processing of LiNbO₃ in different gas composition discharges is a widely used. The ability to form different LiNbO₃ structures with a given qualitative and uniform profile is necessary for various applications, including their use as ridge waveguides [5]. For example, deep (dozens of μm) etching processes are used in the production of LiNbO₃-based MEMS devices. Many works [4, 6, 7] in recent years have been devoted to the etching of the LiNbO₃ in CHF₃, C₄F₈, CF₄, BCl₃ and their mixtures using Ar/N₂/O₂ plasmas. The LiNbO₃ etching process in halogen-containing gas discharges has a relatively low etching rate. The etching rate is usually between 50 and 150 nm min⁻¹ when the power of the inductivity-coupled plasma (ICP) power source is below 1000 W, [2, 3, 8, 9] but it can be increased to 350 nm min⁻¹ by raising the applied power up to 2000 W [10, 11]. It should be noted that the applied power (W_{RF}) is not the only technological parameter that affects the ‘dry’ plasma etching process. Thus, achieving a high etching rate is possible when all the technological process parameters, such as the gas mixture composition (defined by the types of gases and their flow), pressure in the process chamber (P), the bias voltage (U_b) applied to the high frequency (HF) electrode with the sample holder, etc, are optimally combined. The majority of LiNbO₃ plasma etching studies concentrate on these four main parameters, their optimization and their influence on the etching process; however, the temperature of the LiNbO₃ substrate has not previously been considered. Meanwhile, it is well-known that temperature can have a significant effect on the chemical reaction rate. Therefore, the aim of this work was to examine the influence of the LiNbO₃ substrate temperature on the etching rate in SF₆ ICP plasma.



2. Experimental

For this research, the original substrate holder with built-in heater was designed (figure 1). The design process was mostly focused on the uniformity of the temperature distribution on the outer surface. This decision was connected to the fact that crystalline LiNbO_3 is highly sensitive to temperature changes. The uneven heating of a lithium niobate substrate can result in its mechanical destruction. A thermocouple was connected to the central part of the inner space of the holder to control the temperature inside the table. Calibration dependence curves between the temperature of the holder surface and the temperature inside the holder were created. The substrate holder surface temperature was measured with pyrometer Raytek ST20 inserted through the process chamber window.

Z-cut, single-crystal, LiNbO_3 substrates with a 76.2 mm diameter and 370 μm thickness produced by MTI Corporation were used as experimental samples. Two types of masks were used: Cr-Cu-Ni and Cr-Cu-Cr 7 μm thick films with 3×10 mm windows and 10 μm thick films with 0.16×6 mm windows respectively. The experiments were performed in SF_6/O_2 and SF_6/Ar plasmas on an original, specially-designed plasma etching machine with an ICP source. A detailed description of the installation can be found in [12]. The values of the ICP power (W_{RF}), SF_6 flow (Q_{SF_6}), O_2 flow (Q_{O_2}), bias voltage (U_b), and pressure (P) were fixed as 700 W at 6.78 MHz, 10.15 sccm, 3.0 sccm, -50 V and 0.75 Pa, respectively. The etching time was 10 min and the substrate temperature range was 373–598 K.

LiNbO_3 single-crystal wafers were tightly fixed to the substrate holder surface. Afterwards, the processing chamber was evacuated to the required pressure. Then, the substrate was slowly heated to the required temperature and cleaned from contaminants in an Ar plasma for 10 min ($Q_{\text{Ar}} = 21.75$ sccm, $P = 0.7$ Pa, $W_{\text{RF}} = 750$ W, $U_b = 25$ V). Immediately after these preparation procedures, the etching process was started using the mentioned technological parameter values. The etching rate (V_{etch}) was derived from the electron microscopy data (Zeiss Supra 55VP Field Emission Scanning Electron Microscope) using the ratio between the depth of the etching profile and the etching process time.

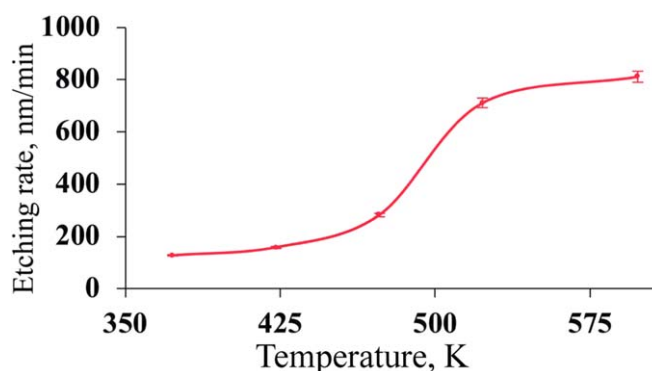


Figure 2. Plot of the etching rate dependency on the temperature.

3. Results and discussion

The obtained etching rate dependence on the substrate temperature is presented in figure 2. The determined dependence has a complex nature with a relatively broad temperature interval. In this region, the variation of the temperature significantly affects the etching rate. Nominally, the curve can be separated into three regions according to its behavior. In the first region (373–423 K), temperature increments provide only small V_{etch} increases (only 31 nm min^{-1} growth) relative to the etching process at room temperature. In the second region (423–523 K), temperature increments of the substrate gives a large increase of the V_{etch} . Here, the V_{etch} grows nearly linearly and reaches 711 nm min^{-1} at 523 K. At even higher substrate temperatures (third region 523–598 K), the rate increment as a function of temperature sharply decreases and provides only 101 nm min^{-1} growth. As a result, our experiments reached a maximum V_{etch} at 598 K of 812 nm min^{-1} , which is double the rates obtained in other studies [10, 11] that were performed with higher source powers but without the additional substrate heating.

In the first region, there is clearly no thermal activation of the reaction between the chemically active particles (CAP) of the plasma and the LiNbO_3 . Here, the etching rate is limited by the CAP delivery conditions as well as the desorption and disposal of the volatile reaction products from the reaction zone. In contrast, the second region corresponds to a thermally activated etching process. In this temperature interval, reactions between the plasma CAP and the LiNbO_3 surface are thermally enhanced and generate the volatile reaction products, which can be easily removed. We assume that in even higher temperatures, the formation of a non-volatile LiF [11] compound occurs and redeposits on the etching surface to block the CAP delivery. As result, this compound does not give further rate increments, even when the temperature of the substrate continues to rise. Thus, the third region shows a limited V_{etch} due to the LiF non-volatile, particle formation.

Microphotographs of the tested samples are presented in figure 3, which shows the results of the LiNbO_3 etching in two different gas mixtures of SF_6/O_2 and SF_6/Ar . Both processes were performed with the mentioned technological conditions and the same temperature of 598 K. Moreover, $Q_{\text{Ar}} = Q_{\text{O}_2} = 3 \text{ sccm}$. The calculation of the etching rate results shows that the replacement of the oxygen with argon in the initial gas mixture adversely affects the etching rate (only 337 nm min^{-1} with argon, which is less than half that when processing with the SF_6/O_2 mixture at 812 nm min^{-1}). In addition, the root mean square roughness is four times greater after etching in the SF_6/Ar gas mixture ($R_{\text{ms}}(\text{SF}_6/\text{Ar}) = 119.37 \text{ nm}$ and $R_{\text{ms}}(\text{SF}_6/\text{O}_2) = 30.52 \text{ nm}$). Our results show that the etching of heated substrate in SF_6/O_2 mixtures provides a better efficiency and better surface quality compared to etching in SF_6/Ar mixtures.

To understand the perspective of the presented plasma etching technique for deep opening formations (dozens of microns), an experiment was performed where the etching time was 270 min. To decrease the previously described non-volatile LiF influence, hydrofluoric acid was used for 30 s after every 90 min in the experiment. The values of the technological parameters were nearly the same as in the experiment with a maximum etching rate value ($W_{\text{RF}} = 700 \text{ W}$, $P = 0.75 \text{ Pa}$, $Q_{\text{SF}_6} = 10.15 \text{ sccm}$, and $Q_{\text{O}_2} = 3 \text{ sccm}$) except for the temperature and bias voltage ($T = 523 \text{ K}$ and $U_b = -80 \text{ V}$). That was done to improve the vertical directionality of the etching process. As can be seen from figure 4, the angle of the side wall after the process was 78 degrees. This occurred because the incoming ions did not have enough energy to continue the vertical, anisotropic etching process during the extended time processing. Thus, the chemical part of the etching process became stronger, which increased the isotropic etching. After 270 min of processing, the etching depth was $113.7 \mu\text{m}$ (figure 4), which corresponds to a 421 nm min^{-1} etching rate, and the etching selectivity ratio of the LiNbO_3 substrate to the Cr-Cu-Cr mask was more than approximately 77. According to the existing LiNbO_3 dry

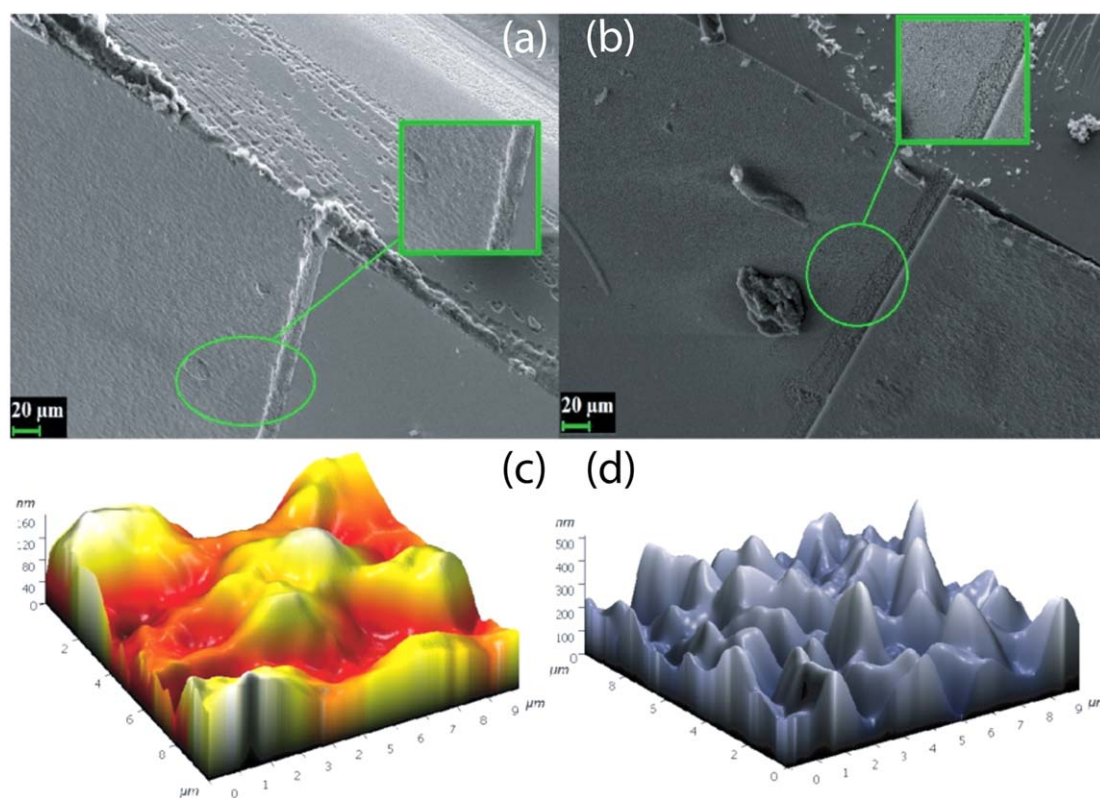


Figure 3. Microphotographs of the etched windows in the Z-cut single-crystal LiNbO₃. (a) -Etching in the SF₆/O₂ gas mixture, (b) etching in the SF₆/Ar gas mixture, (c) atomic force microscopy results after etching in the SF₆/O₂ gas mixture, and (d) atomic force microscopy results after etching in the SF₆/Ar gas mixture.

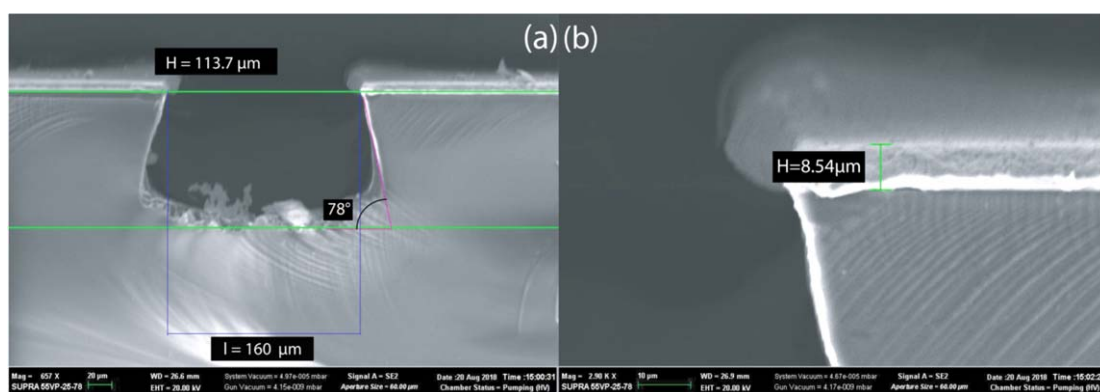


Figure 4. Etched window profile microphotograph after the 270 min process.

etching research, [2, 4, 5] regular etching depths up to 15 μm deep are normal, but our method shows that it is possible to etch up to 113 μm with higher etching rates.

4. Summary and conclusions

In summary, the obtained results show that the thermal enhancement of the substrate is an effective way to increase the etching rate of a single-crystal LiNbO₃. The maximum achieved etching rate was more than 800 nm min^{-1} to an etching depth of 8 μm and a substrate temperature of 598 K with an applied ICP power of less than 1000 W. It was established that etching in a SF₆/O₂ gas mixture achieves higher etching rates with a lower roughness of the etching surface ($R_{\text{ms}} = 30.52 \text{ nm}$ and $V_{\text{etch}} = 812 \text{ nm min}^{-1}$) than etching in the SF₆/Ar gas mixture ($R_{\text{ms}} = 119.37 \text{ nm}$ and $V_{\text{etch}} = 337 \text{ nm min}^{-1}$). An etching rate of 421 nm min^{-1} was observed during the lithium niobate deep etching to a depth of more than 113 μm with a selectivity ratio to the

mask of more than 77. A decrease in the etching rate during the deep etching process was most likely connected to the LiF non-volatile particles that formed during the etching process. The isotropic profile (78 degrees angle of the sidewall) of the etched window is a consequence of the increased chemical etching component of the process.

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